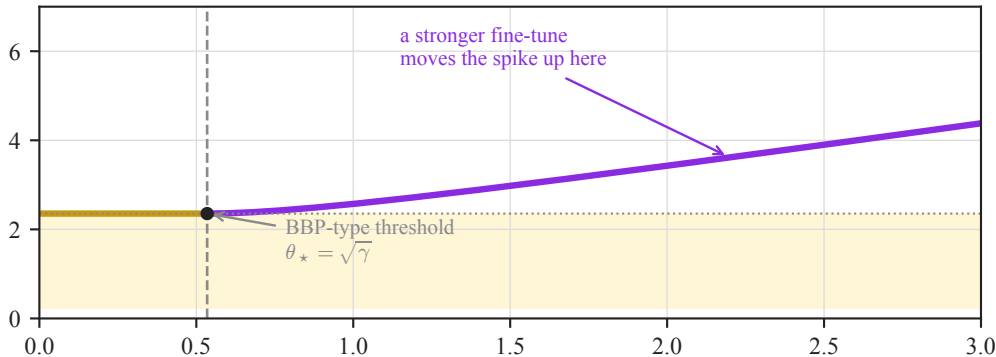


Additive low-rank detectability threshold

observed top eigenvalue of C



BBP-type threshold
 $\theta_* = \sqrt{\gamma}$

a stronger fine-tune
moves the spike up here

- Marchenko--Pastur bulk
- buried: spike inside the bulk
- detached: observable spike